HAT-P-36b

R-0812

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_250319` · created 2026-07-28T12:13:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460753.7802 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1416 +/- 0.0097
Transit depth (Rp/Rs)^2	2.01 +/- 0.27 [%]
Orbital Inclination (inc)	83.44 +/- 2.08
Airmass coefficient 1 (a1)	1.0048 +/- 0.0095
Airmass coefficient 2 (a2)	-0.003 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [627, 239]
Optimal Aperture	3.25
Optimal Annulus	9.36
Transit Duration (day)	0.0878 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1416 ± 0.0097 vs 0.126 (deviation 1.61σ)
Duration fit vs published	0.0878 d vs 0.0925 d (deviation 0.65σ)
Mid-time O–C	-0.7 min
Depth SNR	14.6
Residual scatter	0.95 %

Light curve

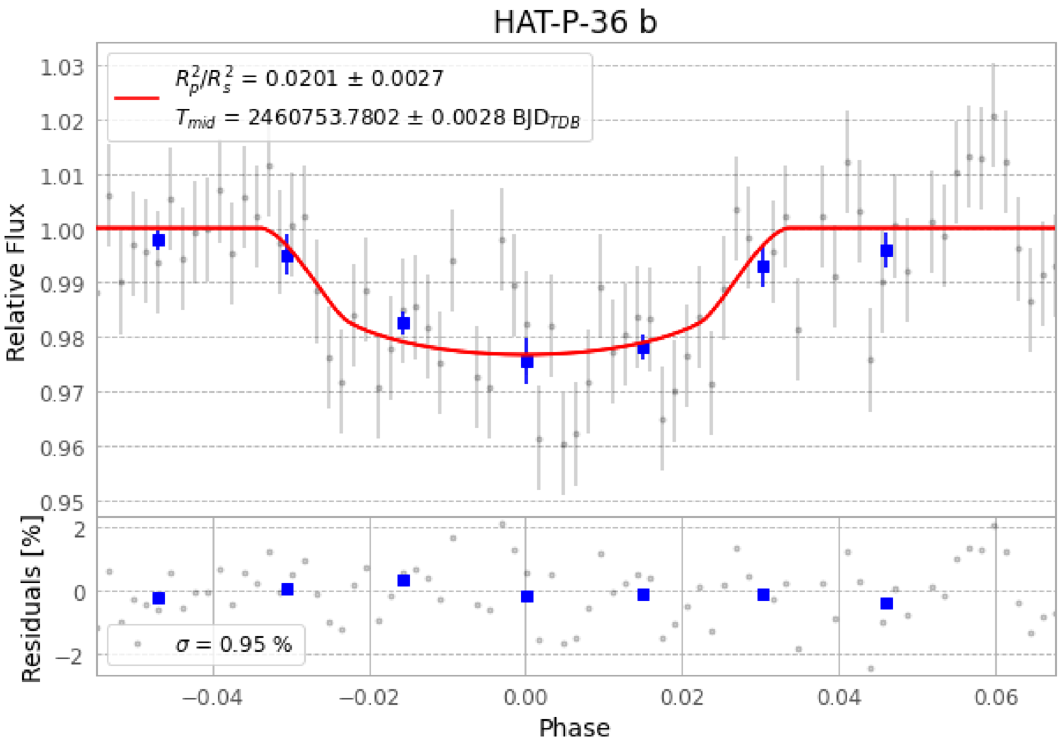

Figure 1 — FinalLightCurve_HAT-P-36 b_2025-03-18.png (SHA-256 738c548fa2cb8629…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 215], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2d357e13641764ea…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

79 FITS frames (52 MB), `HATP-36250319045115.FITS` → `HATP-36250319084515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250319.log (da31917eb5f6…), FinalLightCurve_HAT-P-36 b_2025-03-18.png (738c548fa2cb…), FinalLightCurve_HAT-P-36 b_2025-03-18.csv (c314005c34e7…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)